

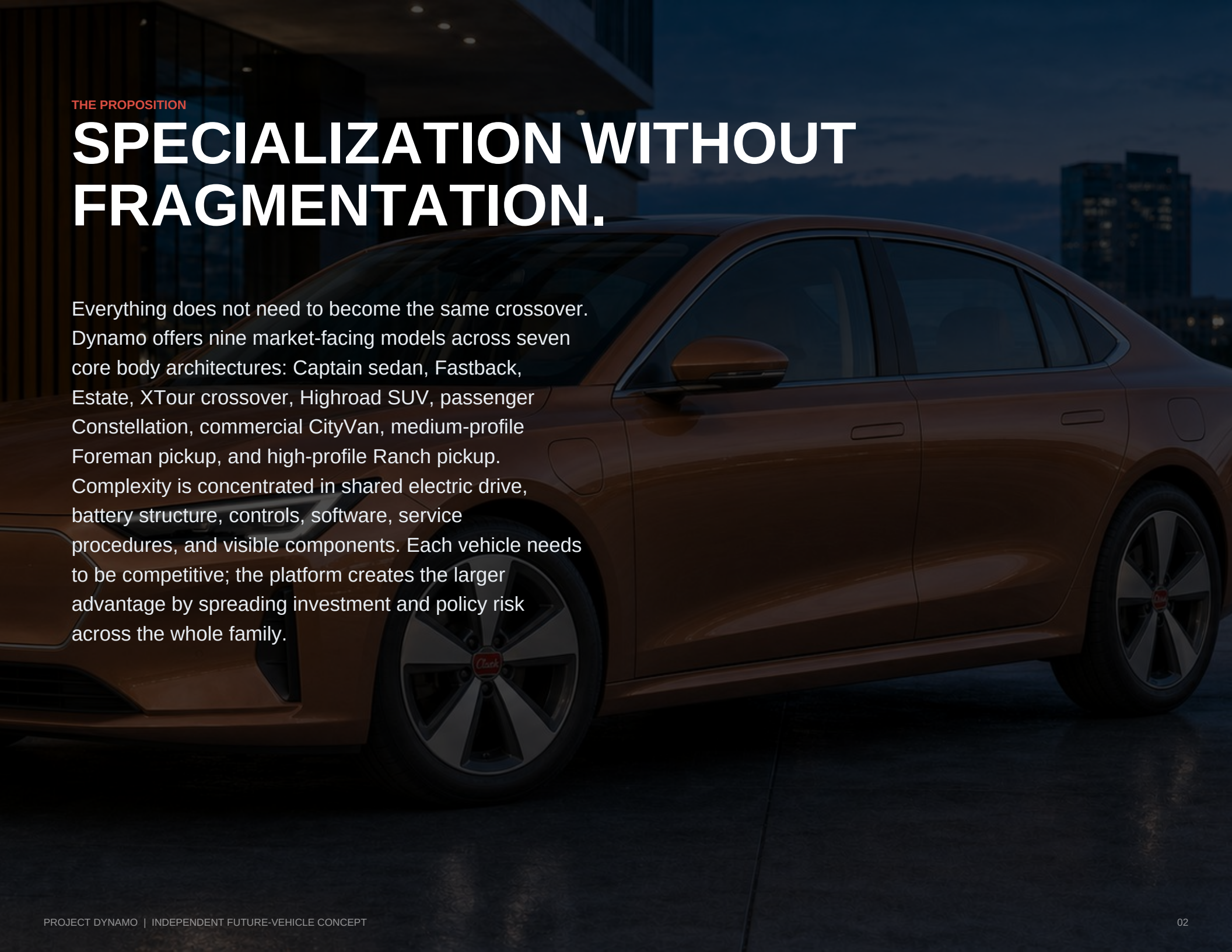


2027 FUTURE VEHICLE CONCEPT

PROJECT DYNAMO

Nine specialized models. One electric-drive
platform. Two energy strategies.

AN INDEPENDENT COGNISINT DESIGN AND PRODUCT-STRATEGY STUDY



THE PROPOSITION

SPECIALIZATION WITHOUT FRAGMENTATION.

Everything does not need to become the same crossover. Dynamo offers nine market-facing models across seven core body architectures: Captain sedan, Fastback, Estate, XTour crossover, Highroad SUV, passenger Constellation, commercial CityVan, medium-profile Foreman pickup, and high-profile Ranch pickup. Complexity is concentrated in shared electric drive, battery structure, controls, software, service procedures, and visible components. Each vehicle needs to be competitive; the platform creates the larger advantage by spreading investment and policy risk across the whole family.



SPECIALIZATION WITHOUT RETOOLING

CAPTAIN. FASTBACK. ESTATE.

Performance derivatives change drive output, brakes, suspension, cooling, wheels, and software—not the platform logic or fundamental assembly sequence.

THE ENGINEERING ANALOGY, MADE PHYSICAL

ELECTRIC TRACTION. FROM RAIL TO ROAD.

Beside a modern North American diesel-electric locomotive, the Dynamo idea becomes tangible: a liquid-fueled prime mover generates electricity while traction motors provide propulsion. Highroad adds a plug-in battery for routine electric miles, power buffering, and regenerative braking.



ONE ARCHITECTURE, TWO DISTINCT ASSIGNMENTS

DYNAMO CONSTELLATION

A worthy successor to the North American passenger minivan.

Shared architecture and tooling. Purposefully different assignments and finishes.



DYNAMO CITYVAN

A worthy successor to the compact commercial van.

The image shows the interior of a car, likely a concept vehicle, with a focus on the driver's side. The seats are upholstered in a light brown leather with a textured pattern. The dashboard is low and horizontal, featuring a large, clear digital display for instruments. The center console has a gear shifter and handbrake. The overall design is minimalist and modern, with a warm, brown color palette. The background shows a city skyline at night through the car's windows.

INSIDE DYNAMO

SIMPLE AND PLUSH

A low horizontal dashboard. Clear instruments. Physical controls for frequent tasks. Warm textiles, broad cushioned surfaces, honest storage, and technology that knows when to disappear.



SHARED VOLUME, DIFFERENT LIVES

CONSTELLATION COMFORT. CITYVAN PURPOSE.

Two correctly aligned, track-mounted second-row captain's chairs sit ahead of one coherent three-place bench.
The commercial configuration gives the same useful volume to organized tools and credible access.

THE COMPLETE 2027 CONCEPT FAMILY

NINE MODELS. FAMILIAR BONES.



CAPTAIN

LOW PROFILE



FASTBACK

LOW PROFILE



ESTATE

LOW PROFILE



XTOUR

MEDIUM PROFILE



FOREMAN

MEDIUM PROFILE



HIGHROAD

HIGH PROFILE



CONSTELLATION

HIGH PROFILE



CITYVAN

HIGH PROFILE



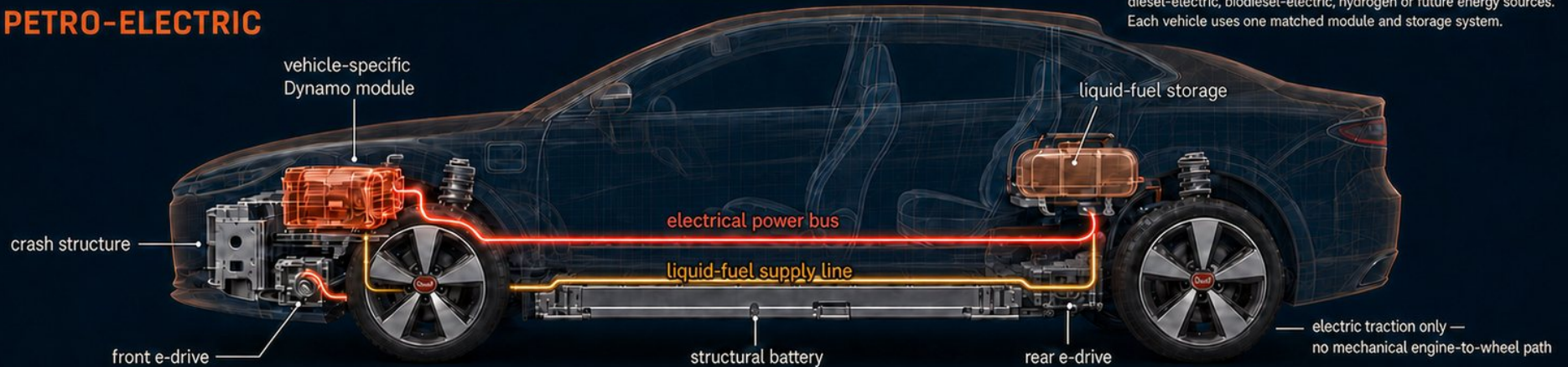
RANCH

HIGH PROFILE

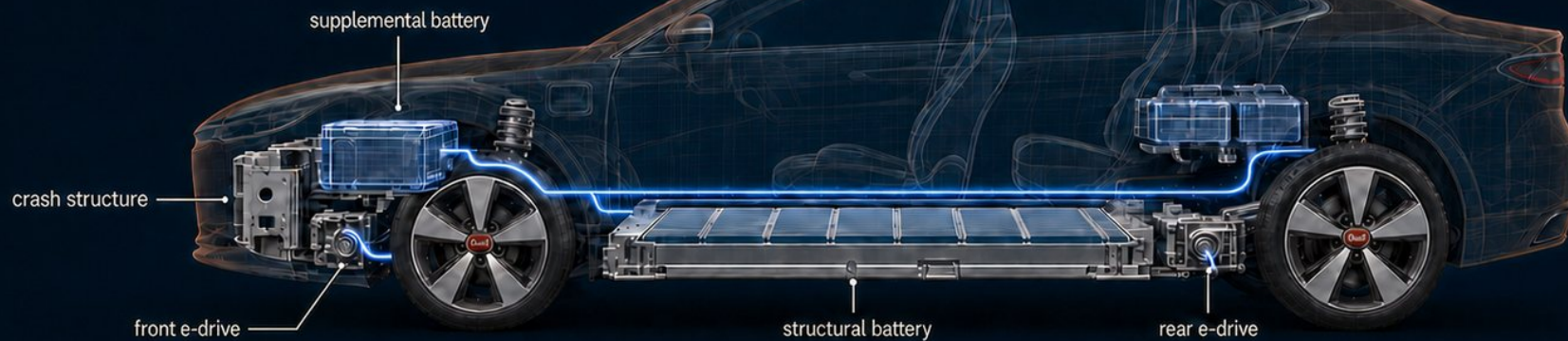
THE ENERGY MODULE CHANGES. THE PROPULSION SYSTEM DOES NOT.

PROJECT DYNAMO

PETRO-ELECTRIC



BATTERY ELECTRIC



ONE ELECTRIC DRIVETRAIN. TWO ENERGY STRATEGIES.

THE DIFFERENCE IS THE PATH TO THE WHEELS.

01 CONVENTIONAL ICE

Fuel > engine > transmission > wheels

Mature and light. Engine speed and road speed remain mechanically coupled.

02 CONVENTIONAL HYBRID

Fuel + battery > engine/motor/transmission > wheels

Efficient and proven, but retains an engine-to-wheel path while adding electric hardware.

03 DYNAMO PETRO-ELECTRIC (PE)

Fuel > Dynamo Module > electricity > motor > wheels

Dynamo's product term for a series hybrid: like a diesel-electric locomotive, a prime mover makes electricity and traction motors alone propel the vehicle.

04 BATTERY ELECTRIC

Grid > battery > motor > wheels

The simplest and lowest-maintenance propulsion system, bounded by stored energy and charging access.

The road vehicle adds a plug-in battery for buffering, regenerative braking, and routine electric miles. Gasoline-, diesel/biodiesel-, or other prime-mover versions are separately engineered module families - not one multi-fuel vehicle.

STOP ASKING AN ICE PLATFORM TO PRETEND IT WAS BORN ELECTRIC.

MANUFACTURING

One body and assembly logic supports both energy strategies.

PARTS

Shared handles, visors, console parts, lamp modules, trim profiles, fascia hardware, wheel families, sensors, and diagnostics.

PACKAGING

Battery, module bay, cooling, crash structure, and cabin designed together.

PROPORTIONS

Long wheelbases, short overhangs, flat floors, and useful cabins.

AWD + ENERGY RECOVERY

Front and rear e-drives give every Dynamo all-wheel drive and traction control; blended brakes recover deceleration energy.

VOLATILITY HEDGE

Policy reversals, fuel-price shocks, and charging delays change the PE/BEV factory mix - not the entire vehicle program.

SIMULATED DIMENSIONS AND PERFORMANCE. GROUNDED ENOUGH TO EVALUATE.

Model	L x W	Weight PE / BEV	Torque	Battery PE / BEV	Gen.	PE mpg	0-60	Top / cruise
Captain	181.8 x 72.2	3,590 / 3,670 lb	295 lb-ft	32 / 72 kWh	80 kW	39	6.1 s	118 / 75
Fastback	179.5 x 72.4	3,630 / 3,710 lb	295 lb-ft	32 / 72 kWh	80 kW	38	6.2 s	116 / 75
Estate	186.2 x 72.6	3,790 / 3,860 lb	315 lb-ft	36 / 78 kWh	85 kW	37	6.4 s	115 / 75
XTour	178.8 x 73.4	3,940 / 4,000 lb	335 lb-ft	38 / 78 kWh	90 kW	34	6.2 s	112 / 75
Highroad	194.0 x 77.4	4,740 / 4,860 lb	500 lb-ft	45 / 105 kWh	120 kW	28	5.8 s	112 / 70
Constellation	196.5 x 76.5	4,480 / 4,560 lb	360 lb-ft	42 / 95 kWh	105 kW	31	7.1 s	105 / 70
CityVan	196.5 x 76.5	4,260 / 4,340 lb	360 lb-ft	42 / 95 kWh	105 kW	32	7.3 s	105 / 70
Foreman	200.5 x 76.8	4,450 / 4,560 lb	420 lb-ft	42 / 90 kWh	105 kW	30	6.3 s	110 / 70
Ranch	211.0 x 79.2	5,060 / 5,200 lb	540 lb-ft	52 / 115 kWh	130 kW	25	5.9 s	110 / 70

PE / BEV values shown in that order. Fuel economy models a gasoline-electric module. Generator-running mpg, acceleration, top speed, sustained level-road cruise, mass, and dimensions are concept targets - not tested or certified claims.

COMPETITIVE ACROSS THE WHOLE USE CASE.

Architecture	Weight	Energy	Battery	Torque	0-60	Annual energy	75k-mi maint.	Strongest case
Efficient ICE	2,875 lb	36 mpg	-	~133 lb-ft	~8.5 s	\$1,167	\$7,575	Low mass, mature infrastructure
Mechanical hybrid	3,208 lb	49 mpg	~1-2 kWh	~232 lb-ft	~6.2 s	\$857	\$7,050	Best gasoline-only efficiency
Captain PE target	3,590 lb	39 mpg gen.	32 kWh	295 lb-ft	6.1 s	\$759	\$6,000	105-mi electric target + rapid refueling
Captain BEV target	3,670 lb	28 kWh/100mi	72 kWh	295 lb-ft	5.9 s	\$571	\$4,575	Lowest energy and maintenance burden

WHERE PE GIVES GROUND

A mechanical hybrid can beat series generation on sustained gasoline highway mpg because it avoids converting mechanical energy to electricity and back.

WHERE BEV LEADS

The full-electric version is simpler, more efficient, and cheaper to maintain because it carries no engine, fuel, exhaust, or emissions hardware.

WHERE PE IS COMPETITIVE

At these energy prices, PE beats the 49-mpg hybrid's annual energy cost above roughly 48% plugged-in electric miles.

THE VEHICLE MUST BE COMPETITIVE. THE PLATFORM CREATES THE LARGER ADVANTAGE.

WHAT STILL HAS TO BE PROVED.

- PE remains more complex and maintenance-intensive than BEV.

- Generator, fuel, exhaust, cooling, and battery can erase mass savings.

- Sustained towing and grades depend on generator output and thermal capacity.

- Occasional engine use creates fuel-aging, catalyst, noise, and emissions challenges.

- A modular bay is not a literal one-for-one volume swap.

- Cost, durability, repairability, certification, and real efficiency require validation.

BOTH HALVES HAVE EXISTED - JUST NOT TOGETHER.

2019: ELECTRIC-ONLY WHEEL DRIVE

Ford's Transit Custom and Tourneo Custom PHEVs drove their front wheels exclusively with an electric motor. A 1.0-liter gasoline engine acted solely as a range extender, with no physical connection to the wheels. The architecture stayed within the Transit Custom family.

2025: ELECTRIC-FIRST PLATFORM

Ford's announced Universal EV Platform applies ground-up electric architecture and new assembly logic to a family of affordable vehicles, beginning with a roughly \$30,000 midsize four-door electric pickup targeted for 2027.

PROJECT DYNAMO IS THE 'YES, AND.'

Combine the Transit's electric-only wheel drive with the new platform's ground-up simplicity, then extend that logic across a complete vehicle family.



ENGINEERED TO BENCHMARKS, NOT DOWN TO A PRICE POINT.

A durable platform preserves more than invested capital: skilled work, supplier capability, service knowledge, useful products, and the trust earned when a company supports what it builds.

Independent concept study. Clark Motor Company is fictional and its red trapezoid logo is a stand-in. Ford Motor Company is named only as a factual precedent and point of comparison. This work was not commissioned, reviewed, sponsored, or endorsed by Ford or any other manufacturer.

All Clark dimensions, performance, costs, ranges, and efficiency figures are simulated design targets, not tested or certified claims. Vehicle and schematic imagery were produced with generative tools under human creative direction.

SOURCES: U.S. DOE AFDC vehicle architecture and maintenance guidance; 2026 Honda Civic specifications; Tesla Model 3 specifications; Ford 2019 Transit Custom PHEV and 2025 Universal EV Platform announcements.

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